

**KARMOY HAUGESUND, Norway**WMO: **014080**Lat: **59.350N**Lon: **5.198E**Elev: **86**StdP: **14.65**Time Zone: **1.00 (EUC)**Period: **07-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 2             | 17.4       | 21.3 | 6.9                           | 7.8 | 21.9 | 11.7 | 10.0 | 25.6 | 38.3                  | 40.2 | 35.1 | 39.1 | 7.6                   | 100  | 0.764 |

**Annual Cooling, Dehumidification, and Enthalpy Design Conditions**

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 9.4                    | 73.5            | 61.8 | 69.6 | 60.6 | 66.5 | 59.0 | 63.8                | 70.6 | 62.0 | 67.2 | 60.4 | 64.7 | 10.8                 | 100  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 61.1                            | 80.7 | 66.1 | 59.6 | 76.3 | 64.3 | 58.3 | 72.9 | 63.0 | 29.1          | 70.9 | 27.8 | 67.1 | 26.7 | 64.9 | 70.3              |

**Extreme Annual Design Conditions**

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |     |            |     |            |     |      |     |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|-----|------------|-----|------------|-----|------|-----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |     | n=20 years |     | n=50 years |     |      |     |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max | Min        | Max | Min        | Max |      |     |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l) | (m)        | (n) | (o)        | (p) |      |     |
| (4) 35.0          | 31.0 | 27.5 | DB                         | 13.4 | 80.7               | 4.7 | 3.9                                                | 10.0 | 83.5       | 7.3 | 85.8       | 4.6 | 88.0       | 1.2 | 90.8 | (4) |
| (5)               |      |      | WB                         | 12.6 | 66.4               | 4.7 | 2.2                                                | 9.2  | 68.0       | 6.4 | 69.3       | 3.8 | 70.5       | 0.3 | 72.2 | (5) |

**Monthly Climatic Design Conditions**

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 46.8   | 35.7  | 35.7  | 38.4  | 43.9  | 49.6  | 54.3  | 59.2  | 59.3  | 55.3  | 48.3  | 42.3  | 38.4 |
|                                                                                  | DBStd       | 9.99   | 6.51  | 5.97  | 4.79  | 4.96  | 5.62  | 4.12  | 4.77  | 3.66  | 4.08  | 4.72  | 6.01  | 6.97 |
|                                                                                  | HDD50       | 2160   | 444   | 399   | 359   | 196   | 72    | 5     | 0     | 0     | 5     | 86    | 234   | 360  |
|                                                                                  | HDD65       | 6679   | 909   | 819   | 824   | 634   | 477   | 323   | 196   | 183   | 291   | 518   | 680   | 825  |
|                                                                                  | CDD50       | 981    | 0     | 0     | 0     | 11    | 61    | 134   | 285   | 288   | 165   | 33    | 4     | 0    |
|                                                                                  | CDD65       | 26     | 0     | 0     | 0     | 0     | 1     | 2     | 16    | 6     | 1     | 0     | 0     | 0    |
|                                                                                  | CDH74       | 112    | 0     | 0     | 0     | 0     | 8     | 9     | 78    | 15    | 2     | 0     | 0     | 0    |
|                                                                                  | CDH80       | 16     | 0     | 0     | 0     | 0     | 0     | 0     | 15    | 1     | 0     | 0     | 0     | 0    |
| Wind                                                                             | WSAvg       | 13.5   | 16.0  | 14.0  | 12.9  | 11.8  | 12.6  | 12.1  | 11.2  | 12.2  | 13.7  | 15.0  | 14.7  | 16.0 |
| Precipitation                                                                    | PrecAvg     | 62.7   | 6.4   | 5.1   | 4.6   | 3.5   | 3.0   | 3.3   | 3.8   | 5.7   | 6.3   | 7.2   | 6.9   | 7.5  |
|                                                                                  | PrecMax     | 75.9   | 11.7  | 9.8   | 9.1   | 6.1   | 5.9   | 6.1   | 6.5   | 10.2  | 10.6  | 11.1  | 12.6  | 12.4 |
|                                                                                  | PrecMin     | 42.1   | 0.4   | 1.1   | 0.1   | 1.6   | 1.0   | 0.9   | 1.8   | 1.4   | 1.8   | 2.4   | 1.4   | 1.5  |
|                                                                                  | PrecStd     | 9.1    | 3.0   | 2.6   | 2.2   | 1.0   | 1.2   | 1.4   | 1.2   | 1.9   | 2.6   | 2.6   | 3.0   | 2.8  |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 47.2  | 46.9  | 50.2  | 64.7  | 74.3  | 73.5  | 82.4  | 75.7  | 71.0  | 62.0  | 55.9  | 50.7 |
|                                                                                  |             | MCWB   | 45.5  | 44.4  | 43.7  | 51.1  | 58.3  | 58.4  | 65.4  | 62.8  | 61.4  | 56.7  | 51.7  | 48.8 |
|                                                                                  | 2%          | DB     | 46.3  | 45.5  | 46.8  | 59.1  | 67.6  | 67.4  | 75.2  | 70.5  | 66.2  | 58.3  | 53.1  | 49.4 |
|                                                                                  |             | MCWB   | 44.5  | 43.7  | 43.4  | 48.1  | 55.4  | 57.7  | 63.6  | 61.1  | 57.4  | 54.0  | 50.3  | 47.7 |
|                                                                                  | 5%          | DB     | 45.3  | 44.6  | 45.3  | 54.9  | 62.1  | 63.6  | 70.3  | 67.5  | 63.2  | 56.3  | 51.4  | 48.2 |
|                                                                                  |             | MCWB   | 43.5  | 43.1  | 42.6  | 46.8  | 54.1  | 56.6  | 62.2  | 60.0  | 57.4  | 52.5  | 48.6  | 46.6 |
|                                                                                  | 10%         | DB     | 44.2  | 43.6  | 44.4  | 51.1  | 58.4  | 60.4  | 66.6  | 64.8  | 61.4  | 54.9  | 50.1  | 46.8 |
|                                                                                  |             | MCWB   | 42.4  | 41.9  | 42.1  | 45.0  | 51.5  | 55.1  | 60.7  | 59.1  | 56.7  | 51.2  | 47.5  | 44.6 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 46.3  | 45.4  | 46.3  | 52.2  | 61.3  | 61.5  | 67.2  | 65.1  | 63.5  | 57.6  | 53.0  | 49.4 |
|                                                                                  |             | MCDB   | 46.6  | 46.2  | 47.8  | 61.4  | 72.6  | 67.9  | 77.4  | 71.0  | 69.1  | 60.3  | 54.1  | 49.9 |
|                                                                                  | 2%          | WB     | 45.3  | 44.2  | 44.6  | 49.5  | 57.1  | 59.1  | 64.8  | 62.6  | 60.2  | 55.6  | 50.9  | 48.2 |
|                                                                                  |             | MCDB   | 45.9  | 44.9  | 45.6  | 55.8  | 64.4  | 65.1  | 72.7  | 67.1  | 63.3  | 57.3  | 52.7  | 49.1 |
|                                                                                  | 5%          | WB     | 44.1  | 43.4  | 43.6  | 47.8  | 54.7  | 57.2  | 63.1  | 61.1  | 58.8  | 54.0  | 49.6  | 46.9 |
|                                                                                  |             | MCDB   | 45.1  | 44.4  | 44.7  | 52.7  | 61.1  | 62.5  | 69.5  | 65.3  | 62.1  | 55.8  | 51.0  | 48.0 |
|                                                                                  | 10%         | WB     | 42.5  | 42.2  | 42.6  | 46.2  | 52.4  | 55.7  | 61.1  | 60.0  | 57.3  | 52.1  | 48.0  | 45.0 |
|                                                                                  |             | MCDB   | 43.8  | 43.4  | 43.9  | 49.8  | 57.2  | 59.6  | 66.1  | 64.0  | 60.5  | 54.3  | 49.5  | 46.5 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 7.6   | 8.0   | 8.0   | 10.6  | 10.5  | 9.1   | 9.0   | 9.4   | 9.1   | 9.0   | 8.0   | 7.6  |
|                                                                                  |             | MCDBR  | 7.4   | 6.4   | 8.1   | 16.5  | 18.0  | 15.0  | 16.8  | 13.2  | 11.9  | 10.2  | 7.1   | 6.9  |
|                                                                                  |             | MCWBR  | 7.2   | 6.1   | 6.1   | 9.2   | 9.3   | 7.8   | 8.3   | 6.9   | 7.2   | 7.7   | 6.8   | 7.6  |
|                                                                                  | 5% WB       | MCDBR  | 7.0   | 6.2   | 6.6   | 12.9  | 16.2  | 12.7  | 14.1  | 10.3  | 9.9   | 7.5   | 7.2   | 6.9  |
|                                                                                  |             | MCWBR  | 7.3   | 6.2   | 5.8   | 8.0   | 8.8   | 7.5   | 7.5   | 6.9   | 7.4   | 7.1   | 7.5   | 7.6  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.308  | 0.333 | 0.352 | 0.381 | 0.368 | 0.363 | 0.377 | 0.376 | 0.360 | 0.339 | 0.311 | 0.286 |      |
|                                                                                  | taud        | 2.167  | 2.278 | 2.338 | 2.309 | 2.420 | 2.466 | 2.445 | 2.444 | 2.474 | 2.410 | 2.241 | 2.105 |      |
|                                                                                  | Ebn at Noon | 148    | 213   | 247   | 260   | 274   | 277   | 269   | 260   | 245   | 211   | 148   | 110   |      |
|                                                                                  | Edh at Noon | 15     | 22    | 28    | 34    | 33    | 32    | 32    | 30    | 25    | 19    | 14    | 11    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 87     | 266   | 643   | 1186  | 1639  | 1754  | 1536  | 1183  | 753   | 364   | 123   | 54    |      |
|                                                                                  | RadStd      | 13     | 45    | 84    | 108   | 178   | 207   | 146   | 108   | 65    | 46    | 16    | 7     |      |

**Historical Trends**

|                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                         |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| Regional (34 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | -211  | N/A   | N/A   |

Nomenclature: See separate page